DIGITAL FORENSICS INVESTIGATION REPORT

Windows Disk Forensics — Insider Threat & Data Exfiltration

Case Name	CyberDefenders — Hunter
Case Number	2026-001
Examiner	Tej Chaute
Date	20/03/2026
Platform	CyberDefenders
Evidence File	Hunter.ad1
Hash (SHA1)	eb5c7d1cd094772f42b1a6acda38217433dd4190
File Size	606 MB
Classification	Training

1. Case Summary

A forensic investigation was conducted on a Windows 8.1 disk image (Hunter.ad1) provided via the CyberDefenders Hunter challenge. The objective was to analyse user activity, identify suspicious behaviour, and determine whether any malicious or unauthorised actions were performed by the system user.

Analysis revealed that the suspect system was used for network reconnaissance, covert external communication, data staging, and attempted exfiltration. Evidence further confirmed the use of anti-forensic tools — including a file shredder and disk encryption software — in an attempt to conceal malicious activity. The user demonstrated clear and deliberate intent across multiple artifact types.

2. Evidence Examined

Item	Filename	Hash (SHA1)	File Size	Format
001	Hunter.ad1	eb5c7d1cd094772f42b1a6acda38217433dd4190	606 MB	AD1

3. Tools Used

Tool	Version	Purpose
Autopsy	4.22.1	Primary disk image analysis — file browsing, keyword search, artifact extraction
FTK Imager	4.7.3.81	AD1 image mounting and file extraction
Registry Explorer	2.1.0.0	Offline registry hive analysis — SYSTEM, SOFTWARE, NTUSER.DAT
EvtxECmd	1.5.2.0	Windows Event Log (.evtx) parsing and export
WinPrefetchViewer	1.3.7.0	Prefetch file parsing — execution history and run counts
DB Browser for SQLite	3.13.1.0	Skype main.db database analysis
JLECmd (EZ Tools)	1.5.1	Jump List analysis — application execution and file access history

4. Key Findings

Each finding below is supported by at least one artifact. Path, timestamp, and source tool are documented for every item.

4.1 System Information

Field	Value
OS	Windows 8.1
Computer name	4ORENSICS
User account	hunter
System SID	S-1-5-21-2489440558-2754304563-710705792
IP address	10.0.2.15 (DHCP)
Timezone	UTC-07:00 (Pacific Time, DST applied)
Last login	2016-06-21 01:42 UTC

4.2 Artifact Table

#	Artifact Type	Location / Path	Timestamp (UTC)	What It Proves
1	Registry — computer name	SYSTEM\ControlSet001\Control\ComputerName\ComputerName	N/A	Computer name identified as 4ORENSICS
2	Registry — IP address	SYSTEM\CurrentControlSet\Services\Tcpip\Parameters\Interfaces	N/A	System IP address: 10.0.2.15 (DHCP assigned)
3	Registry — DHCP lease	SYSTEM\CurrentControlSet\Services\Tcpip\Parameters\Interfaces	2016-06-21 02:24:12	DHCP lease time confirmed via Unix epoch value in registry

#	Artifact Type	Location / Path	Timestamp (UTC)	What It Proves
4	Registry — SID	SOFTWARE\Microsoft\Windows NT\CurrentVersion\ProfileList	N/A	System SID: S-1-5-21-2489440558-2754304563-710705792
5	Registry — OS version	SOFTWARE\Microsoft\Windows NT\CurrentVersion	N/A	Operating system confirmed as Windows 8.1
6	Registry — timezone	SYSTEM\ControlSet001\Control\TimeZoneInformation	N/A	Timezone: UTC-07:00 (Pacific Time with DST)
7	Event Log — logon count	Windows\System32\winevt\Logs\Security.evtx (Event ID 4624)	N/A	User 'hunter' logged in 3 times total — confirms regular system access
8	Event Log — last logon	Windows\System32\winevt\Logs\Security.evtx (Event ID 4624)	2016-06-21 01:42:00	Last successful login by user 'hunter' — establishes session start time
9	Prefetch — Zenmap execution	C:\Windows\Prefetch\ZENMAP.EXE-56B17C4C.pf	2016-06-21 12:08:13	Zenmap (GUI for Nmap) executed — confirms network scanning tool was run by user
10	Nmap XML output — scan end time	C:\Users\hunter\Desktop\nmapscan.xml	2016-06-21 05:12:09	Port scan completed at this timestamp — 1000 ports scanned using Nmap v7.12
11	Nmap XML output — open ports	C:\Users\hunter\Desktop\nmapscan.xml	2016-06-21 05:12:09	Open ports identified on target: 22 (SSH), 80 (HTTP), 9929, 31337 — potential attack surface mapped
12	Nmap XML output — tool version	C:\Users\hunter\Desktop\nmapscan.xml	2016-06-21 05:12:09	Nmap version 7.12 confirmed — specific version recorded for attribution
13	Skype DB — conversation	C:\Users\hunter\AppData\Roaming\Skype\hunterehpt\main.db	2016-06-21 09:08:36	User communicated with external account 'linux-rul3z' via Skype — confirms contact with external party
14	Skype DB — exfil agreement	C:\Users\hunter\AppData\Roaming\Skype\hunterehpt\main.db	2016-06-21 09:08:36	Conversation confirms agreement to use TeamViewer for remote access and data exfiltration — establishes intent
15	Skype DB — linked email	C:\Users\hunter\AppData\Roaming\Skype\hunterehpt\main.db	2016-06-21 09:08:36	Skype account linked to ehptmsgs@gmail.com — identifies the suspect

#	Artifact Type	Location / Path	Timestamp (UTC)	What It Proves
16	Email attachment — network diagram	C:\Users\hunter\Documents\Outlook Files\backup.pst	2016-06-21 13:13:43	Network diagram image found in Outlook PST — later deleted, suggesting evidence removal after external communication
17	User document — exfil research	C:\Users\hunter\Documents\Ryan_VanAntwerp_thesis.pdf	2016-06-21 09:40:07	PDF discussing data exfiltration techniques — confirms user researched exfiltration methods
18	Install log — Crypto Swap	C:\Program Files\Uninstall.log	2016-06-21 11:44:40	Disk encryption tool installed — anti-forensic capability, potential use for hiding data before exfiltration
19	Registry — USB device 1	SYSTEM\ControlSet001\Enum\USBSTOR	2016-06-21 01:53:14	First USB storage device connected (Serial: 07B20C03C80830A9) — potential exfiltration vector
20	Registry — USB device 2	SYSTEM\ControlSet001\Enum\USBSTOR	2016-06-21 02:01:59	Second USB storage device connected (Serial: AAI6UXDKZDV8E9OU) — second potential exfiltration device
21	Installed app — BCWipe	C:\Program Files\Jetico\BCWipe	2016-05-23 09:40:22	Jetico BCWipe file shredder installed — presence before the incident date indicates pre-planned anti-forensic capability
22	Prefetch — BCWipe execution	C:\Windows\Prefetch\BCWIPE.EXE-36F3F2DF.pf	2016-06-21 17:32:35	BCWipe executed 5 times — repeated secure deletion confirms deliberate attempt to destroy evidence
23	Downloaded file — Burp Suite	C:\Users\hunter\Downloads\Burpsuite_free_v1.7.03.jar	2016-06-21 11:00:58	Burp Suite web proxy tool present and executed — confirms capability to intercept and manipulate network traffic
24	Jump List — Tor Browser	C:\Users\hunter\AppData\Roaming\Microsoft\Windows\Recent\CustomDestinations\ (AppID: aa28770954eaeaaa)	2016-06-21 12:08:50	Tor Browser executed — confirms deliberate use of anonymisation tool to evade network monitoring
25	Email artifact — Pictures.7z	User email history / mail database	N/A (undated)	Compressed archive 'Pictures.7z' sent as email attachment — confirms data exfiltration attempt via email
26	Shellbags — staging folder	NTUSER.DAT — Shellbags (Explorer artifacts)	N/A (undated)	Folder C:\Users\Hunter\Pictures\Exfil created and accessed —

#	Artifact Type	Location / Path	Timestamp (UTC)	What It Proves
				confirms deliberate data staging prior to exfiltration
27	Recycle Bin — deleted image	C:\$Recycle.Bin\	2016-06-21 09:31:09	File 'ws_Small_cute_kitty_1920x1200.jpg' deleted — initial cleanup activity after accessing sensitive content
28	Prefetch directory count	C:\Windows\Prefetch\	N/A (historical)	174 valid Prefetch files identified (excluding zero-byte and duplicates) — confirms extensive system usage history

5. Event Timeline

Reconstructed chronological sequence of events based on artifact timestamps. All times are in UTC. Undated artifacts (Shellbags, email exfil attempt) are listed separately below the main timeline as their sequence within the incident cannot be precisely established.

Timestamp (UTC)	Artifact Source	Event Description
2016-05-23 09:40:22	Installed application	Jetico BCWipe (file shredder) installed — pre-incident, indicates pre-planned anti-forensic capability
2016-06-21 01:42:00	Security.evtx (Event ID 4624)	User 'hunter' last logged into the system successfully
2016-06-21 01:53:14	Registry — USBSTOR	First USB device connected (Serial: 07B20C03C80830A9)
2016-06-21 02:01:59	Registry — USBSTOR	Second USB device connected (Serial: AAI6UXDKZDV8E9OU)
2016-06-21 02:24:12	Registry — DHCP	System obtained IP address 10.0.2.15 via DHCP — network connection established
2016-06-21 05:12:09	Nmap XML output	Port scan completed — 1000 ports scanned, open ports 22, 80, 9929, 31337 identified using Nmap v7.12
2016-06-21 09:08:36	Skype main.db	User communicated with 'linux-rul3z' via Skype — agreed to use TeamViewer for remote access and data exfiltration
2016-06-21 09:08:36	Skype main.db	Skype account linked to ehptmsgs@gmail.com — suspect identity confirmed
2016-06-21 09:31:09	Recycle Bin	Image file deleted — initial file cleanup activity begins
2016-06-21 09:40:07	Documents	Ryan_VanAntwerp_thesis.pdf accessed — user researched data exfiltration techniques
2016-06-21 11:00:58	Downloads	Burp Suite v1.7.03 JAR executed — web traffic interception tool activated
2016-06-21 11:44:40	Install log	Crypto Swap (disk encryption) installed — anti-forensic tooling expanded

Timestamp (UTC)	Artifact Source	Event Description
2016-06-21 12:08:13	Prefetch — ZENMAP.EXE	Zenmap executed — network scanning performed against target
2016-06-21 12:08:50	Jump List — Tor Browser	Tor Browser launched — anonymisation tool used to evade network monitoring
2016-06-21 13:13:43	Outlook PST	Network diagram identified in email data — sensitive asset accessed, later deleted
2016-06-21 17:32:35	Prefetch — BCWIPE.EXE	BCWipe executed 5 times — repeated secure deletion of files to destroy evidence

Undated artifacts — sequence within the incident cannot be precisely established from available timestamps:

Artifact Source	Description
Email — Pictures.7z	Compressed archive sent as email attachment — confirms data exfiltration attempt via email channel
Shellbags — Exfil folder	Folder C:\Users\Hunter\Pictures\Exfil created and browsed — data staging confirmed prior to exfiltration
Prefetch directory	174 valid Prefetch files found — confirms extensive application execution history across the investigation period

6. Conclusion

The forensic investigation determines that user 'hunter' (email: ehptmsgs@gmail.com) conducted a deliberate and coordinated sequence of malicious activities on the Windows 8.1 system (computer name: 4ORENSICS) on 2016-06-21, with anti-forensic tools pre-installed as early as 2016-05-23.

Initial activity centred on network reconnaissance. Nmap v7.12 (via Zenmap, artifact #9) was used to scan 1000 ports on a target host, identifying open ports 22, 80, 9929, and 31337 (artifacts #10–12). This was preceded by two USB devices being connected (artifacts #19–20), which represent likely exfiltration vectors. The user then made contact with an external party ('linux-rul3z') via Skype at 09:08 UTC (artifact #13), during which both parties explicitly agreed to use TeamViewer for remote access and data exfiltration (artifact #14).

Evidence of exfiltration intent is confirmed through multiple independent sources: a research PDF on exfiltration techniques (artifact #17), a dedicated staging folder created at C:\Users\Hunter\Pictures\Exfil (artifact #26), and an email attachment named Pictures.7z sent externally (artifact #25). Additional offensive tools — Burp Suite (artifact #23) and Tor Browser (artifact #24) — confirm capability to intercept traffic and anonymise activity.

Anti-forensic behaviour is strongly evidenced by the pre-installed Jetico BCWipe file shredder (artifact #21), which was executed 5 times on the day of the incident (artifact #22), alongside the installation of Crypto Swap disk encryption software (artifact #18) and the deletion of a sensitive network diagram from Outlook data

(artifact #16). The combination of BCWipe's repeated execution and Crypto Swap installation immediately after external communication strongly indicates deliberate destruction of evidence.

Based on the totality of evidence, it is concluded that the user intentionally performed network reconnaissance, coordinated with an external attacker to plan data theft, staged and exfiltrated sensitive data, and actively used anti-forensic tools to conceal their actions. The attack was premeditated, as evidenced by BCWipe being installed nearly a month before the incident date.

7. Examiner Notes

This investigation was conducted retrospectively as part of a structured DFIR learning path. Notes were partially recorded during the live investigation and completed after challenge submission.

Key methodological observations from this case:

- Prefetch analysis (artifacts #9, #22) was the most effective technique for proving tool execution, even in the absence of the original executables. WinPrefetchViewer accurately decoded run counts and last execution timestamps.
- Skype database analysis (main.db via DB Browser for SQLite) proved essential for establishing intent — the explicit exfiltration agreement in chat logs would be highly significant in a real criminal or HR investigation.
- Registry USBSTOR keys (artifacts #19–20) identified removable media without the physical devices being present — a critical technique in insider threat investigations.
- Shellbags (artifact #26) revealed a staging folder that would not have been visible through standard file browsing — reinforcing the value of parsing NTUSER.DAT rather than relying on Autopsy's file tree alone.
- Correlating timestamps across different sources and accounting for the UTC-07:00 timezone required careful attention. All timestamps in this report have been normalised to UTC.

If this were a real-world investigation, recommended next steps would include: full recovery and hash verification of deleted files from unallocated space, deep inspection of the Pictures.7z email attachment contents, network traffic analysis to confirm TeamViewer connections, memory forensics for volatile evidence of running processes at time of incident, and correlation with server-side logs to confirm exfiltration success.